Durability of hybrid FRP reinforcing bars in concrete structures exposed to marine environments

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Abstract: To solve the brittle failure of existing FRP rebar, we developed hybrid FRP rebar that has more than 3% strain and plastic deformation like a steel reinforcement. So, the developed hybrid FRP rebar has emerged as one of the most promising and affordable solution to the brittle failure problem of existing FRP rebar in concrete structures.

In this study, the durability of hybrid FRP rebar exposed to marine environments was evaluated. The performance degradation of three types of hybrid FRP rebar and one type of GFRP rebar were investigated after exposure to marine environment. Laboratory simulated a marine environment with solutions of NaCl, Na_2SO_4, and $CaCl_2$, and repeated dry/wet and freeze/thaw cycles after immersion in salt water. Performance assessment was based on tensile strength, short beam tests, and bond tests. All test results showed high residual strength under the all condition. The experimental results confirm the desirable resistance of hybrid FRP rebar to marine environment.

Keywords: durability; fibre-reinforced polymer; FRP; hybrid; marine.

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1 Introduction

Recently, the civil structures are increasingly constructed in marine region. The reinforced concrete was consists of concrete to strong in compression and steel reinforcement to strong in tension. So, it was known that the RC structures were durable structure due to the excellent adhesion between concrete and steel reinforcement. However because of short of river sand, sometimes the structures were constructed to used sea sand. These structures were damaged and the embedded steel reinforcement was corroded by salt. The salt was the biggest degradation factor of RC structures (Clear et al., 1995). This problem was particularly serious in concrete structures that were in direct contact with water such as marine structure and concrete bridges but satisfactory solution has not yet been found (Won and Park, 2006). Therefore many studies were performed to prevent damage from salt.

And also steel reinforcement corrosion of marine structures which it was always exposured environmental degradation by salt was the important factor that determined the persisting period of structures. Concrete structures has disadvantage that it was decreased strength and increased volume change and drying shrinkage by salt which supplied to external or internal. Especially, concrete expansion and porous structure by salt was occurred the crack. The crack was acted like a corridor that was supplied salt in to the concrete and steel reinforcement. Because of the supplied salt, the steel reinforcement was corroded and expanded. The corroded steel reinforcement decreased adhesion with concrete body. Therefore it was decreased mechanical performance of concrete structures.

But when exposed the marine environment, the performance degradation factor of hybrid fibre-reinforced polymer (FRP) was different from steel reinforcement. It is maybe occur the fracture by chemical reaction different from corrosion. So, it is important to evaluate the durability of FRP rebar in marine environment (Benmokrane and Masmoudi, 1996; Vijay, 1999).

FRP rebar has emerged as one of the most promising and affordable solution to the corrosion problems of steel reinforcement in structural concrete. FRP rebar in concrete structures may be used as a substitute of steel reinforcement for the cases in which aggressive environment produce high steel corrosion or lightweight is an important factor in design and construction. But, reinforcement of FRP bar for concrete usually has linear elastic behaviour up to tensile failure. For safety reasons, a certain plastic deformation and an elongation greater than 3% at maximum load is usually required for steel reinforcement in concrete structures (Fujita and Hamada, 1996). The same is desirable for FRP rebars. Without plastic deformation, a small increase in load can cause catastrophic collapse without any warning (Tamuzs and Tepfers, 1995). Thus, the current FRP rebars are not suitable for concrete reinforcement where a large amount of plastic deformation prior to collapse is required. Therefore we developed hybrid FRP rebar that has more than 3% strain and plastic deformation like a steel reinforcement (Won et al., 2007).

The desirable performance of hybrid FRP rebar was obtained from the integrated design based on the material hybrid and geometric hybrid concepts. Hybrid FRP rebar was produced by pultrusion, braiding and filament winding techniques. In case of pultrusion process is making a core of FRP, it is consist of reinforcement supply, resin impregnation, heated die and puller. The braiding process is making a sleeve to increase stain, compressive and shear strength. At last, filament winding process is making a rib at FRP surface to increase bonding property with concrete body. Through this process, we manufactured hybrid FRP rebar.

The tensile test results showed that the hybrid FRP rebar stress-strain curves are linearly elastic with a definite yield point followed by plastic deformation.

The durability of hybrid FRP rebar, when used in a marine-based structure, was examined under accelerated degradation conditions in artificial seawater with repeated wet/dry and freeze/thaw cycles. Pull-out tests were preformed to evaluate bonding with the concrete body, and shear strength and tensile performance tests were conducted to evaluate the long-term durability against seawater of concrete reinforced with these hybrid FRP rebars.

2 Experimental

2.1 FRP rebar

The hybrid FRP rebars used in this study was made of three types of fibre, E-glass, aramid and carbon. The core of all of hybrid FRP rebar was made of carbon fibre and held together with a vinylester resin binder. In addition, all of hybbrid FRP rebar has a sleeve to increase stain, compressive and shear strength and a rib at FRP surface to increase bonding property with concrete body. The component ratio of each hybrid FRP rebar used in this study is shown in Table 1. Tests were performed in parallel with Aslan 100 glass fibre-reinforced polymer (GFRP) rebar (Hughes Brothers, USA) for comparison. The GFRP rebar contained approximately 70% glass fibre by volume held together with a vinylester resin binder. Figure 1 shows the morphology of FRP rebars used in this study. The diameter of GFRP and hybrid (C) was 6mm and the other hybrid FRP was 4 mm.

Table 1 Mix proportions of hybrid FRP rebar

Type of hybrid FRP rebar	*Fibre types (percent of volume fraction, %)*		
	E-glass	*Aramid*	*Carbon*
Hybrid (A)	-	73.0	27.0
Hybrid (B)	81.0	-	19.0
Hybrid (C)	51.8	34.6	13.6

Figure 1 The morphology of used FRP rebars (see online version for colours)

2.2 Exposure environments

2.2.1 Chemical environments

The sea water was contained variety chemical component. Among these, we were chosen the chemical componts which the biggest factor of degradation. To create an accelerated reproduction of chemical degradation in seawater, specimens were exposed to solutions of 3% NaCl at 60°C, 10% Na_2SO_4 at 20°C, and 4% $CaCl_2$ at 20°C for 50 days.

2.2.2 Physical environments

Marine concrete structures experience repeated drying and wetting cycles due to waves and tides. To reproduce these effects, the hybrid FRP rebar was dried for 24 h at approximately 60°C, and then immersed in 3% NaCl at 20°C for 24 h. This cycle was repeated 25 times. Seasonal changes were also taken into consideration. The hybrid FRP rebar was immersed for 24 h in 4% soluton of NaCl at approximately 20°C, and then frozen for 24 h at –10°C. This process was also repeated 25 times.

2.3 Test methods

2.3.1 Interlaminar shear stress test

The most FRP rebar has low Interlaminar shear stress (ISS). It mean that frcture occurred easily between fibres which conbined with polymer. Because of the fracture at layer between fibres was determined by polymer resin which comparatively weaker than fibre. In this study, ISS testing was performed to evaluated the shear property of hybrid FRP rebar after exposured degradation environments in accordance with ASTM D 4475 using a 50-kN capacity universal testing machine (UTM) and a strain velocity of 1.3 mm/min (ASTM D 4475, 2002). Figure 2 shows the interlaminar shear test set-up state. The length between the surpported points was 3 times of diameter. Load was translated to a intelaminar shear stress according to the following formula (ASTM D 4475, 2002):

$$S = 0.849\frac{P_{\max}}{D^2} \tag{1}$$

where S represents the intelaminar shear stress in MPa, $P_{\max}$ is fracture load in kN, and D is the diameter in mm.

Figure 2 Interlaminar shear test set-up (see online version for colours)

2.3.2 Tensile test

Tensile tests were performed according to the standards given by the ACI 440 committee with a 250-kN UTM and loading velocity of 5 mm/min (ACI 440.1R-03,2003) (Castro and Carino, 1998). Figure 3 shows the tensile test set-up state and the specimen was prepared in accordance with ACI 440. In the standards given by the ACI 440 committee , the test length of FRP rebar was regulated longer than 40 times of rebar diameter and the minimum langth was more than 100 mm. And the anchor length was regulated more than 250 mm at least. Therefore we manufactured specimen which has test length 300 mm and 250 mm anchor length. So, total length of specimen was 800 mm.

Figure 3 Tensile test set-up (see online version for colours)

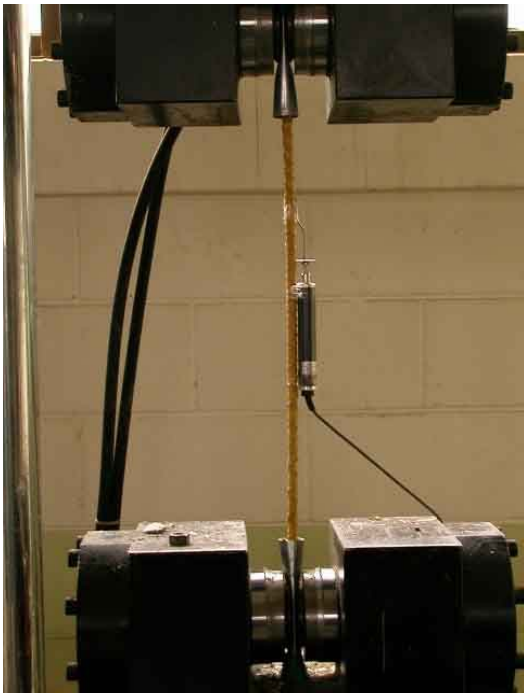
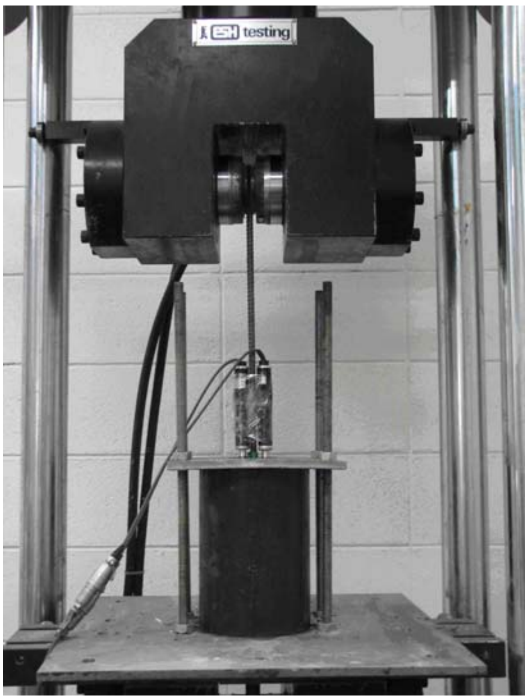


Figure 4 Pull-out test set-up



2.3.3 Pull-out test

A direct bond test was performed as suggested by the ACI 440 committee. The concrete surrounding the rebar was manufactured cylinder specimen Φ 150 × 300 mm and cured for 28 days. The embedded depth of FRP rebar in concrete was 5times of diameter. The direct bond test was performed to used 250-kN UTM with a loading velocity of 5 mm/min). Figure 4 shows the direct bond test set-up state. Load and strain were translated to a bond strength according to the following formula (ACI 440.1R-03, 2003; Nanni et al., 1996; Nanni, 1996):

$$\tau = \frac{P_{\max}}{2\pi DL} \tag{2}$$

where τ represents the bond strength in MPa, P_{max} is the maximum bond load in kN, D is the diameter in mm, and *L* is the embedded depth in mm.

3 Results

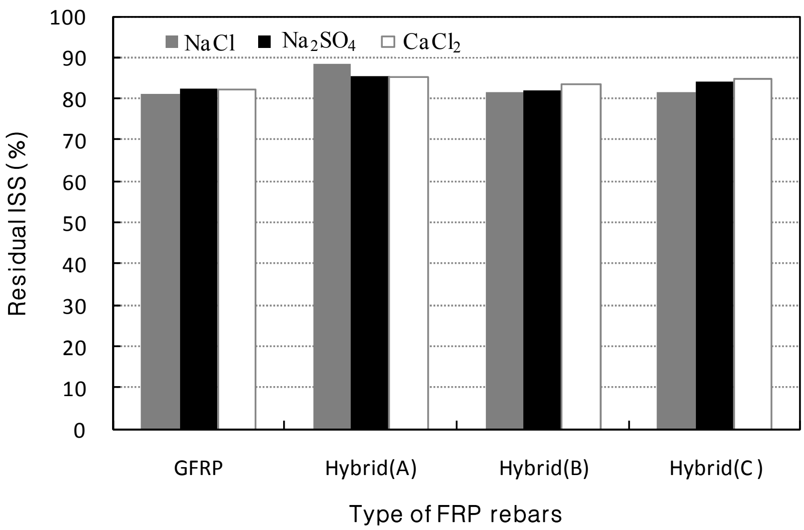
3.1 ISS test

3.1.1 Chemical environments

The shear strength of GFRP not exposed to any of the chemical environments described above was 532.2 MPa; the corresponding values for hybrid FRP rebars (A), (B), and (C) were 1052.4, 831.7, and 932.6 MPa, respectively.

Figure 5 shows the ISS value of FRP rebar after exposure to the various chemical environments. For all three of the salt solutions, the hybrid FRP retained over 80% of its initial value, indicating no major effects on the mechnical properties of the structure despite direct exposure to seawater. For comparison, the GFRP rebar exhibited a residual ISS of 80.5%, also indicative of no major effects.

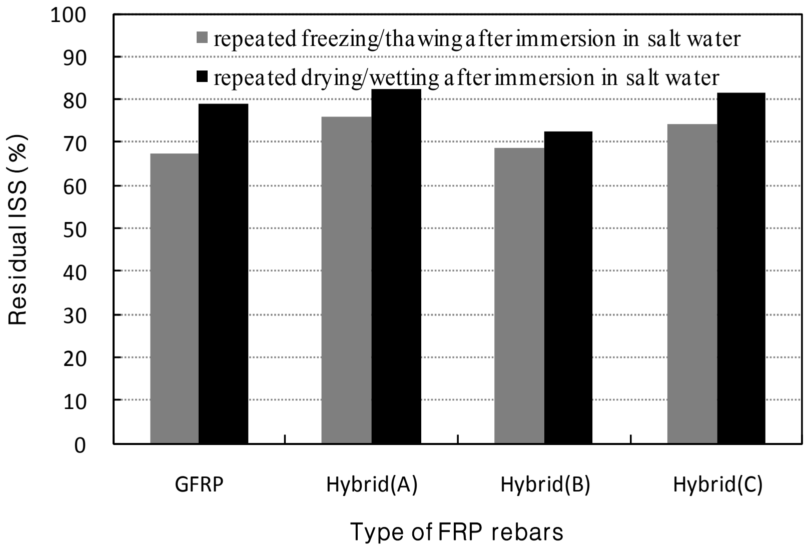
Figure 5 ISS test results after exposure to chemical environments



3.1.2 Physical environments

Figure 6 shows the ISS test results after exposured physical degradation environments. As results, the hybrid FRP rebar (B) and GFRP rebar, which both contain glass fibre as the main constituent, exhibited only 70% of their initial ISS values following repeated drying and wetting with seawater. Thus, slight performance degradation may occur with repeated drying and wetting cycles. Conversely, hybrids (A) and (C) retained over 80% of their initial ISS values, indicating excellent resistance to the effects of repeated drying and wetting with seawater.

Figure 6 ISS test results after exposure to physical environments



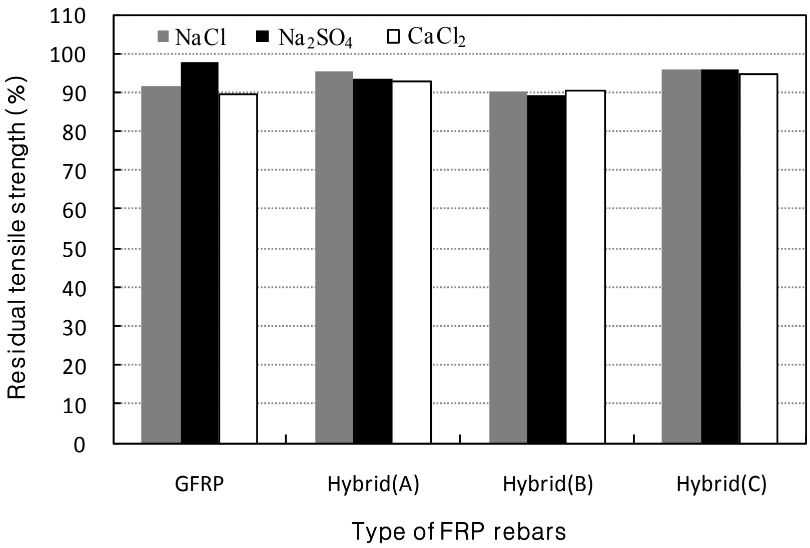
Hybrid FRP rebars (A), (B), and (C) exhibited 76.1%, 68.8%, and 74.4%, respectively, of their initial ISS values following freeze/thaw testing. The GFRP rebar had a residual ISS of 67.6%. The FRP rebars incorporating glass fibres were more susceptible to wear with resisdual ISS values under 70%. This is because the glass fibre is less durable than carbon or aramid fibres. However, the evaluation standard of 65% residual ISS was satisfied in all cases.

3.2 Tensile tests

3.2.1 Chemical environments

Figure 7 shows that the residual tensile strength for all of the evaluated rebars was over 80% of each initial value under all conditions. In particular, the hybrid FRP rebars maintained approximately 90% of their initial tensile strength, indicating a better chemical resistance than the GFRP rebar. In addition, the hybrid B shows less than the other hybrid FRP rebar. It is indicating that aramid and carbon fibre has a better chemical resistance than the glass fibre. But, unlike the GFRP rebar, all of the hybrid FRP rebar maitained its ductility.

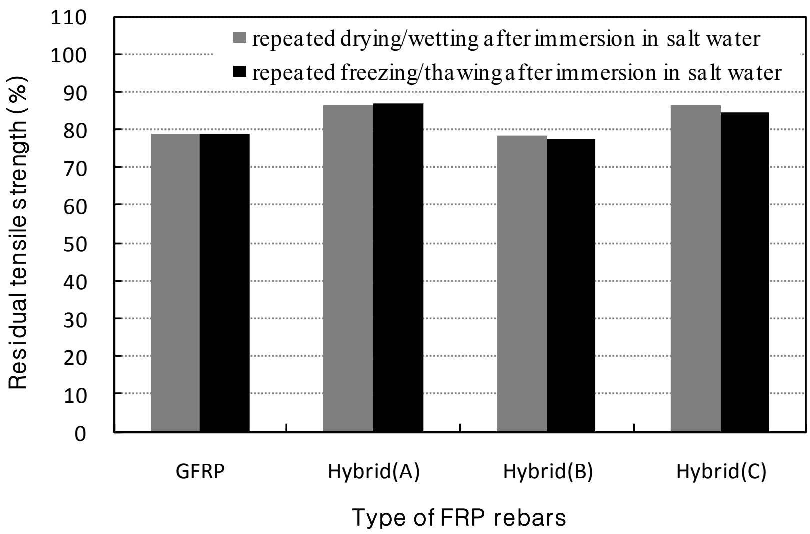
Figure 7 Tensile test results after exposure to chemical environments



3.2.2 Physical environments

Figure 8 shows that hybrid FRP rebar (B) and GFRP rebar, which are primarily glass fibre, exhibited 80% of their initial tensile strengths following repeated wet/dry cycles with seawater. Hybrid FRP rebars (A) and (B), which contain aramid and carbon fibre, respectively, maintained 85% of their tensile strengths, indicating excellent performance and wear resistance.

Figure 8 Tensile test results after exposure to physical environments



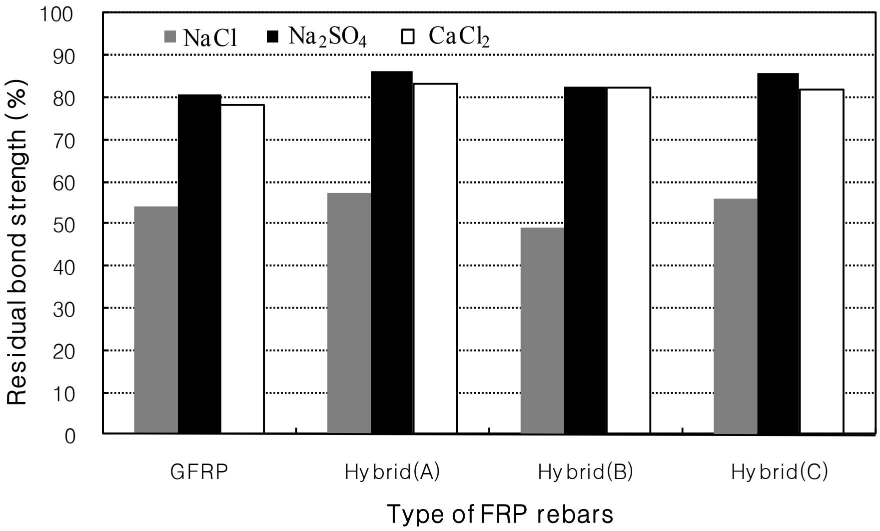
The residual tensile strength of hybrid FRP rebars (A), (B), and (C), and the GFRP rebar following several freeze/thaw cycles were 87%, 77%, 84%, and 79% respectively. This indicates a slight degradation in the durability of FRP rebars based on glass fibres.

3.3 Pull-out tests

3.3.1 Chemical environments

The bond strength of unexposed GFRP was 174 kgf/cm^2. Hybrid FRP rebars (A), (B), and (C) exhibited bond strengths of 280 kgf/cm^2, 235 kgf/cm^2, and 260 kgf/cm^2, respectively. The residual bond strength of FRP rebars exposed to the various chemical environments is shown in Figure 9. The FRP rebar exposed to Na_2SO_4 and $CaCl_2$ solutions exhibited a residual bond strength over 80%, indicating excellent durability. In contrast, the FRP rebar exposed to NaCl solution maintained a residual bond strength of only 50%. Hybrid FRP rebar (B) had the lowest residual bond strength of approximately 48%. Thus under these circumstances, performance degradation, defined in this case as a drop in pull-out resistance, appears to be due to the glass fibre surface ribs that can be easily damaged by friction with the concrete substrate upon pull out (Tamuzs et al., 1999).

Figure 9 Pull-out test results after exposure to chemical environments

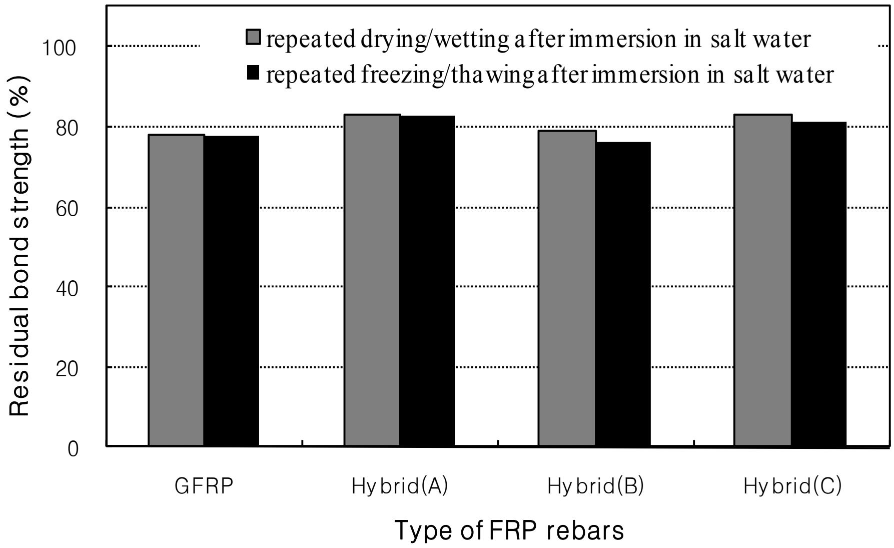


3.3.2 Physical environments

The residual bond strength of FRP rebar exposed to repeated dry/wet and freeze/thaw cycles is shown in Figure 10. Hybrid FRP rebars (A) and (C) showed residual bond strengths of over 80%, indicating minimal degradation of durability. Even the GFRP rebar and hybrid FRP rebar (B), with residual bond strengths of 78% and 79%, respectively, did not show great degradation. The freeze/thaw cycles produced similar results with hybrid rebars (A) and (C), exhibiting residual bond strengths of over 80%.

The GFRP rebar and hybrid FRP rebar (B) had residual bond strengths of 77.5% and 76.4%, respectively, indicating relatively low bond degradation.

Figure 10 Pull-out test results after exposure to physical environments



4 Conclusions

To evaluate the durability of hybrid FRP rebar exposed to marine environments, five types of accelerated environments were reproduced. To evaluate the mechnical propertiess of FRP rebar exposed to each environment, the performance was reviewed with shear, tensile stength, and bond tests. A summary of important findings is as follows:

1 Hybrid FRP rebar (A) exhibited excellent residual strength following exposure to all of the accelerated environments. This is likely a result of its carbon and aramid fibre composition, which was held together with a vinylester resin binder.

2 The strength of hybrid FRP rebar (B) was slightly compromised following several dry/wet and freeze/thaw cycles in artificial seawater. These processes appear to affect the glass fibre. Thus, this hybrid should be considered similar to GFRP rebar.

3 Hybrid FRP rebar (C) showed excellent durability after exposure to the accelerated marine environments. As with hybrid (A), its durability is likely due to the combination of carbon and aramid fibres with the vinylester resin, although hybrid (C) does contain glass fibre.

4 The GFRP rebar exhibited significant performance degradation to 50–60% of its initial bond strength following a 50-day exposure in 3% NaCl solution at 60°C. This is likely due to the presence of a surface layer composed of glass fibres.

5 The hybrid FRP rebars showed excellent residual bond strength following exposure to NaCl, Na_2SO_4, and $CaCl_2$ environments, indicating no great effects under such conditions.

6 Repeated dry/wet and freeze/thaw cycles had minimal effect on the hybrid FRP rebars, as indicated by residual bond strengths of approximately 80%.

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References

ACI 440.1R-03 (2003) 'Guide for the design and construction of concrete reinforced with FRP bars', p.440, American Concrete Institute Committee, Farmington Hills, MI, USA.

ASTM D 4475 (2002) 'Standard test method for apparent horizontal shear strength of pultruded reinforced plastic rods by the short beam method', ASTM International, West Conshohocken, PA, USA.

Benmokrane, B. and Masmoudi, R. (1996) 'FRP C-bar as reinforcing rod for concrete structures', *Proceeding Of Advanced Composite Materials in Bridges and Structure 2nd International Conference*, Montreal, Quebec.

Castro, P.F. and Carino, N.J. (1998) 'Tensile and non-destructive testing of FRP bars', *Journal of Composites for Constructions*, Vol. 2, No. 1, pp.17–28.

Clear, K.C., Hartt, W.H., McIntyre, J. and Lee, S.K. (1995) *Performance of Epoxy-Coated Reinforcing Steel in Highway Bridges*, National Cooperative Highway research Program, Report 370, TRB, National Academy Press, Washington, D.C.

Fujita, A. and Hamada, H. (1996) 'Tensile properties of braided composite rod', *Proceeding 1st International Conference on Composites in Infrastructures, ICCI 96*, pp.451–465, Tucson, Arizona.

Nanni, A. (1996) *Fiber-Reinforced Plastics (FRP) Reinforcement for Concrete Structures Properties and Applications*, pp.167–188, Elsevier Pub. Co. Inc., NY.

Nanni, A., Bakis, C.E. and Terosky, J.A. (1996) 'Smart pseudo-ductile, reinforcing rods for concrete: manufacture and test', in *Proceeding 1st International Conference On Composites in Infrastructures, ICCI 96*, pp.95–108, Tucson, Arizona.

Tamuzs, V. and Tepfers, R. (1995) 'Ductility of non-metallic hybrid fiber composite reinforcement for concrete', *Proceeding of the Second International RILEM Symposium*, Ghent, Belgium, pp.18–25.

Tamuzs, V., Tepfers, R., Apinis, R. and Modniks, J. (1999) 'Pull-out, flexural rotation capacity and creep tests using hybrid composite rods and CFCC rods for reinforcement in concrete', Work, No. 32, pp.99–102, Institute of Polymer Mechanics Latvian Academy of Sciences Aizkraukles 23, Ríga, LV-1006.

Vijay, P.V. (1999) 'Aging and design of concrete members reinforced with GFRP bars', PhD thesis, Department of Civil Engineering West Virginia University, Morgantown, West Virginia, USA.

Won, J-P. and Park, C-G. (2006) 'Effect of environmental exposure on the mechanical and bonding properties of hybrid FRP reinforcing bars for concrete structures', *Journal of Composite Materials*, Vol. 40, No. 12, pp.1063–1076.

Won, J-P., Park, C-G. and Jang, C-I. (2007) 'Tensile fracture and bond properties of ductile hybrid FRP reinforcing bars', *Polymers and Polymer Composites*, Vol. 15, No. 1, pp.9–15.